

Anthropomorphism in AI-Enabled Banking Chatbots: A Systematic Review and Research Agenda

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Abstract

Banking chatbots are becoming more friendly. They greet customers by name, offer a bit of conversational filler, and sometimes even have a name of their own. The logic behind all of this is simple enough: make the chatbot more human-like and customers will trust it more and be more satisfied. This paper tests that idea. It reviews the empirical literature on anthropomorphism in AI-enabled banking and financial services chatbots, drawing on work published between 2007 and 2026 in marketing, human-computer interaction, and information systems. What emerges is a picture more complicated than the industry assumption. Anthropomorphic cues – visual, linguistic and conversational – do tend to increase perceived humanness, and through it, affective trust and satisfaction. But there is a counter-current in the same literature that gets far less attention. In a utilitarian, high-stakes environment such as banking, too much anthropomorphism, or anthropomorphism not supported by genuine competence, can erode perceived reliability and generate frustration or anger, rather than warmth. This paper brings both strands together into a calibration framework that tries to explain when human-like design helps and when it backfires specifically in banking. It concludes with a research agenda tackling four largely unaddressed gaps: longitudinal measurement, context-specific calibration thresholds, differences between voice and text modalities, and the need to revisit older findings given how generative AI has changed what a chatbot is capable of. The review is intended as a focused companion to more general AI-enabled service-quality research, not as a replacement for it.

Keywords: Anthropomorphism, Chatbots, Banking, Trust, Customer Satisfaction, Human-Computer Interaction, Systematic Review.

1. Introduction

Ask a bank customer what they really want from a chatbot and the answer is often prosaic: a fast, accurate answer to a routine question. But that's not what banks have been building lately. Increasingly they're creating systems that are meant to feel like someone is on the other end. A nickname. A figure. Chit-chat. An apologetic tone when something is wrong. None of this is accidental. This is anthropomorphism in action, our tendency to attribute human-like qualities, intentions or emotions to something that's not human. Over the past few years, quietly it has become one of the more consequential, and more debated, design choices in AI-enabled banking. Why is this important? Because the assumption that drives a lot of

this design work, that more human-like automatically means more trusted and more satisfying, only holds up half the time. There is a lot of research on when this holds and when it does not, spread across marketing journals, HCI conferences, and information-systems outlets, but not much of this research talks to each other in a banking-specific way. This paper seeks to fill that gap. It reviews empirical research on anthropomorphism in banking and adjacent financial services chatbots, synthesizes what is known about its causes and its effects on trust and satisfaction, and attempts to map out when anthropomorphism helps and when it subtly backfires. And that last bit is really the point of the exercise. One scope notes in front. The paper is more limited than a full review of the quality of AI-enabled banking services would be. It identifies anthropomorphism and explores this in greater depth than a multi-dimensional service-quality framework realistically could for any single dimension. In Section 2, the theoretical background is explored, ranging from social psychology to banking-specific HCI research. Section 3 presents the findings by theme. Section 4 applies these themes to a calibration framework. Section 5 offers a research agenda. Section 6 discusses implications for theory and practice. Section 7 concludes.

2. Theoretical Foundations of Anthropomorphism

2.1 The Three-Factor Theory of Anthropomorphism

Much of the contemporary research on anthropomorphism can be traced back to Epley, Waytz and Cacioppo's (2007) three-factor theory published in *Psychological Review*. The argument is that people attribute human qualities to non-human agents (animals, brands, machines or, here, banking chatbots) under three conditions. First, if it is cognitively available to the agent and appears to be like human knowledge. Second, when one is motivated to explain or predict what the agent will do next, what the authors call reflectance motivation. Third, when someone does not have enough social connection to actual humans, the motivation is when they seek it elsewhere. A follow-up study by Epley, Waytz, Akalis, and Cacioppo (2008) directly tested the motivational side and found that loneliness and a stable need for control each independently motivated person to anthropomorphize more. In other words, anthropomorphism is not just a feature of good interface design. Partly it's the property of the person using the interface.

That individual differences angle doesn't get much airtime in the marketing and HCI literature covered in Section 3, but it matters for banking specifically. The tendency of a customer to anthropomorphize a chatbot may have less to do with how the chatbot was built and more to do with things like loneliness, a need for control or a bad prior experience with human staff at the branch. A design tweak that reliably boosts perceived humanness in a lab study might behave very differently once it encounters this sort of real-world variation, which lab studies aren't really designed to catch.

2.2 Anthropomorphism in Human-Computer Interaction

Alongside the psychology, there's a design-focused HCI literature asking a more practical question: which interface cues make something feel human? Go and Sundar (2019), still one of the most cited papers in this space, broke chatbot humanization down into three cue types. Visual (avatar, appearance). Identity (a name, some implied or disclosed human origin). And conversational (natural language flow, well-timed responses, small talk). Conversational cues came out on top by a clear margin, with visual and identity cues doing comparatively little on their own. That's a useful finding for banking chatbot design, since it suggests appearance matters a lot less than how the thing talks. And talk is cheaper to fix than visual design.

There's an older, related idea worth bringing in here too: Mori's (1970) uncanny valley. The basic claim is that people's comfort with a human-like agent rises as it gets more human-like, but only up to a point. Push past that point without reaching full realism and comfort drops sharply into unease, recovering only once the agent becomes indistinguishable from an actual person. Mori was writing about robots, and the idea later got extended to computer-generated faces, but the underlying logic, that more human-like isn't always better and can even reverse, shows up again in a different form in the banking-specific finding covered later in Section 3.4.

2.3 Anthropomorphism as a Formal AI Service-Quality Dimension

The clearest bridge between this psychological and HCI work and banking service-quality research is Noor, Rao Hill, and Troshani's (2022) AISQUAL scale in the European Journal of Marketing. Built from interviews with both AI service-agent users and domain experts, AISQUAL treats anthropomorphism as one of six validated dimensions of AI service-agent quality, alongside efficiency, security, availability, enjoyment, and contact, and finds it an unusually strong predictor of perceived quality. What AISQUAL really does is take anthropomorphism out of the controlled-experiment world and turn it into something a customer survey can measure. That's the same move this paper's author relies on elsewhere, in a separate doctoral study modelling anthropomorphism as one dimension within a broader AI-enabled Service Quality (AISQ) framework. This review exists to do the opposite: take that one dimension and sit with it long enough to see what a seven-dimension instrument must skip past.

3. Thematic Synthesis of Findings

3.1 Design Cues That Drive Perceived Anthropomorphism

The visual-identity-conversational typology of Go and Sundar (2019) introduced in Section 2.2 remains relevant, but more recent work has drilled down into specific cue categories with greater relevance to banking. A meta-analytic structural equation modelling study (2025) that brought together chatbot-adoption research across industries found that linguistic anthropomorphism, meaning conversational register, first-person phrasing, an appropriately informal tone, was the cue category most clearly linked to affective, emotion-based satisfaction. It significantly outperformed visual and identity cues, supporting and extending Go and Sundar's earlier work on conversational cues, but in the domain of satisfaction rather than humanness perception alone. A 2025 ScienceDirect study of banking voicebot acceptance in Malaysia took things one step further, splitting anthropomorphism from two related but distinct constructs: social presence (the feeling that you are interacting with something socially real) and media richness (how many communications cues the channel can carry). Across 448 surveyed adults, all three predicted usefulness, enjoyment, and trust. But not in the same way.

3.2 Anthropomorphism's Effect on Trust: A Dual Pathway

One finding recurs often enough in this literature to merit separate attention. Trust appears to be affected by anthropomorphism in two distinct ways, not one. A qualitative study of 28 participants in Pakistan and China (Scientific Reports, 2026) found that affective trust was driven by human-like interaction and emotional connection, while a separate, more cognitive trust pathway was driven by perceived reliability and system competence (accuracy, transparency, responsiveness, data security). The Malaysian voicebot study above revealed a similar pattern: anthropomorphism and social presence influenced both cognitive and emotional trust, whereas media richness mainly influenced usefulness and enjoyment rather than trust per se. The downstream consequences of this were quantified in an Indian banking-sector study (PMC, 2023). Its hypothesized antecedents accounted for 38.6% of the variance in banking chatbot trust, which

accounted for 9.9% of the variance in customer attitude, 11.4% in behavioural intention, and 13.6% in user satisfaction. The chain is not huge, but statistically real, and placed particularly in the same Indian banking context upon which the parent doctoral research of this paper is based.

What this means practically is that a bank cannot assume a single design fix will move both trust pathways at the same time. You could make the chatbot seem warmer, more human, increasing affective trust, without changing its actual transactional reliability and leaving cognitive trust as is. Or you could significantly improve backend accuracy and consistency and not have the felt sense of rapport shift at all for the customer. Some banking survey instruments still treat trust as a single undifferentiated number, risking the obscuring of which of these two pathways a given AI investment is moving down.

3.3 Anthropomorphism's Effect on Satisfaction

A study focusing on the effect of chatbot anthropomorphism on trust and satisfaction through perceived value found that anthropomorphic design increased customers' sense of value, and it was this increase in value, not anthropomorphism directly, that predicted satisfaction. This is an important distinction for any study including the parent doctoral research that underlies this paper that considers anthropomorphism and perceived value as related but separate constructs within the same model. Eren (2021) showed that trust in service provided by chatbots, along with bank's reputation, was a critical determinant of overall satisfaction in banking. This suggests that the effectiveness of anthropomorphism in banking may be somewhat dependent upon the reputational context in which it is used, rather than being equally effective across contexts.

3.4 The Double-Edged Sword: When Anthropomorphism Backfires in Banking

Perhaps the most important finding to emerge from this review and the one with the clearest practical bite is the fact that banking is a high-stakes, highly utilitarian service seems to turn some of the general anthropomorphism-benefits story on its head. Furthermore, a recent study by Youn and Cho (2023) found that customers in utilitarian settings such as banking actually prefer chatbots with less anthropomorphism, as they are seen as being more reliable and efficient than their more human-like counterparts, a direct counterexample to the idea that human-like design is simply always better. Adam, Wessel, and Benlian (2021) arrived at a similar finding: customers are more likely to comply with plain, less anthropomorphic chatbots that they perceive as more transparent and functional, especially when the goal is simply to get the task done rather than to have a conversation.

In a widely cited study, and for good reason, Crolic, Thomas, Roggeveen, and colleagues (2021) extend this. Excessive or poorly calibrated anthropomorphic design can actually provoke anger when the human-like design cues of a chatbot raise expectations of human-level competence that the actual performance does not meet. It's basically the uncanny valley of 2.2, replayed in a service context, but here the unease comes from a mismatch between seeming and actual competence, as opposed to physical appearance. Mozafari, Weiger, and Hammerschmidt (2022) add a wrinkle: disclosing that a chatbot isn't human helps or hurts trust, depending on how much interaction on the interaction. Disclosure costs more in high-stakes situations like a disputed transaction, than in low-stakes ones like a balance check. On top of this, Cheng et al. found an almost paradoxical result. When their identities were revealed, trust was lower in chatbots who tried to appear empathetic, but higher in chatbots who tried to appear just friendly. In other words, the same anthropomorphic choice can be directed in opposite directions, depending on the precise way in which it is expressed.

Table 1 lays this tension out directly, contrasting the conditions where literature finds anthropomorphism helpful against the conditions where it tends to backfire.

Condition	Anthropomorphism Tends to Help	Anthropomorphism Tends to Backfire
Task type	Social/relational interactions (e.g., financial advisory, complaint handling)	Purely transactional tasks (e.g., balance check, fund transfer)
Stakes of the interaction	Low-stakes, low-risk queries	High-stakes queries (disputed transactions, fraud reports)
Calibration of competence to presentation	Human-like presentation matched by accurate, competent performance	Human-like presentation paired with errors or failures (uncanny/anger response)
Identity disclosure	Friendly (not empathetic) persona with disclosed AI identity	Empathetic persona with disclosed AI identity in high-stakes contexts
Cue type	Conversational and linguistic cues (tone, phrasing)	Overinvestment in visual/avatar cues without matching conversational quality

3.5 Anthropomorphism in the Indian Banking Context

The India-specific evidence here, relative to the broader global literature above, is still thin but what there is is directly useful for this paper's parent doctoral research. The most directly relevant Indian quantitative evidence is the PMC (2023) study in Section 3.2, which places chatbot trust within the broader disruption narrative in Indian banking. The Pakistan-China qualitative study (Scientific Reports, 2026) is not India-specific, but it is a closer South Asian cultural comparison than the Western European and North American samples that dominate the anthropomorphism literature. The central gap around which the research agenda of Section 5 is built is this lack of India-specific work, especially anything that tests the double-edged-sword tension from Section 3.4 within an Indian customer population.

4. An Integrative Calibration Framework

Taken together, the findings in Section 3 don't reduce neatly to 'more is better' or 'less is better.' What they point to instead is a calibration framework, where anthropomorphism's net effect on trust and satisfaction hinges on three things lining up: where the task sits on a utilitarian-to-relational spectrum, how much is riding on that particular interaction, and whether the chatbot's actual competence matches the human-likeness it is projecting. Get those three aligned, a relational task, modest stakes, competence that matches the presentation, and Sections 3.1 through 3.3 back a positive chain running from anthropomorphism through trust to satisfaction. Get them misaligned, a purely transactional task, high stakes, or a gap between presented and actual competence, and Section 3.4 points to a flat, negative, or actively counterproductive relationship instead.

There is a measurement implication buried in here too, relevant to this paper's parent doctoral study among others. A single Likert-scale anthropomorphism score, averaged across a mixed bag of banking tasks and customers, can hide effects that cancel each other out: strongly positive in relational, low-stakes moments, flat or negative in transactional, high-stakes ones. That cancellation only becomes visible if anthropomorphism's relationship with trust and satisfaction is tested separately by task type or interaction stakes, rather than treated as constant across a respondent's whole undifferentiated banking experience.

5. Research Agenda

Building on the calibration framework above, and the specific gaps highlighted throughout Section 3, four areas stand out as priorities for future work.

- Measurement over time. Nearly all the studies reviewed here are cross-sectional, capturing the effect of anthropomorphism at one point in time in a customer's relationship with a chatbot. We don't really know yet if the positive effects in Sections 3.1–3.3 or the negative ones in Section 3.4 get stronger, weaker, or flip entirely when you have repeated interactions with the same chatbot over months or years.
- Context specific calibration limits Section 3.4 shows that there is a double-edged sword, but none of the studies studied identify where the line between helpful and harmful anthropomorphism lies for a given task-and-stakes combination. It could indeed estimate that threshold rather than only showing it is there, with experimental work varying anthropomorphism systematically across a controlled matrix of task type and stakes within one banking population.
- Voice vs. text modality effects. The Malaysian voicebot study in Section 3.1 suggests that voice-based anthropomorphism may operate through somewhat different mechanisms, media richness, social presence, than the text-based mechanisms most of this literature examines. As Indian banks venture deeper into voice-based AI, particularly for less digitally literate or vernacular-language customers, this distinction merits dedicated attention rather than an afterthought.
- Re-evaluated under the conversational capability of the generative-AI era. Much of the foundational work covered in Sections 2 and 3, Go and Sundar (2019), Epley et al. (2007), predates the conversational sophistication of today's generative AI. The cue categories and calibration thresholds identified in earlier, less capable chatbot generations, may or may not apply when a chatbot can hold a genuinely open-ended, contextually coherent conversation, rather than the more scripted systems most of these studies examined. Given how fast the underlying technology is moving, this is an increasingly urgent question.

A fifth gap should be mentioned. As discussed in Section 3.5, there is still very little India-specific evidence that directly tests the double-edged-sword tension discussed in Section 3.4. Thus, a systematic study investigating the effect of anthropomorphism on trust and satisfaction across transactional vs relational banking tasks, among the Indian customer population, covering both digitally native and less digitally engaged segments, would directly address that gap. It would be a complement rather than a duplicate to the broader AI-enabled service-quality research, including the parent study of this paper, that sees anthropomorphism as one dimension of several and not the whole story.

6. Implications for Theory and Practice

For theory, what this review mainly contributes is pulling together a real, empirically supported tension: anthropomorphism helps in some banking contexts and hurts in others, a pattern that has so far only shown up piecemeal, scattered across separate studies that rarely get read next to each other. The calibration framework in Section 4 gives future hypothesis-testing something to organize around besides the simplified assumption that anthropomorphism just straightforwardly benefits satisfaction. For practice, the clearest takeaway is that making a chatbot feel more human should not be treated as a design goal to be maximized everywhere. Banks would do better to calibrate anthropomorphic design against the specific task at hand, leaning into warmer, more conversationally rich anthropomorphism for relational and advisory interactions, while keeping high-stakes transactional moments like fraud disputes or large fund

transfers plainer and more transparently AI-presented, since the evidence here suggests customers reward reliability and transparency over rapport in exactly those moments.

7. Conclusion

Anthropomorphism has moved from a peripheral design flourish to a formally recognized dimension of AI service quality. What this review confirms is that as a design choice it is not uniformly good or uniformly dangerous. It is contingent, and its impact depends on whether a chatbot's depicted human-likeness is appropriate for the task at hand and what is truly at stake in that task. Banking is in a weird place here, with real relational ambitions like financial advisory and long-term customer relationships, combined with high-stakes transactional moments that are unusually high-stakes, so this sort of conditionality probably matters more here than in the more low-stakes retail and hospitality spaces that most of the broader anthropomorphism literature is based on. The research agenda in Section 5 is intended to turn this conditionality from something people have noticed into something precisely measured, and the calibration framework in Section 4 gives banks an interim, theory-based, foundation for making anthropomorphic design decisions in the meanwhile, rather than falling back on the still common, and only partially justified, assumption that more human-like is simply better.

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